M378K: November 13th, 2026.

Maximum Likelihood Estimation.

Likelihood.

Def'n. $Y_1, \dots, Y_n \sim D$ (discrete) w/ an unknown parameter θ

The likelihood f'tion is

$$L(\theta; y_1, \dots, y_n) = p^\theta(y_1) \cdot p^\theta(y_2) \cdot \dots \cdot p^\theta(y_n) \leftarrow$$

w/ p^θ the pmf of D.

Example. Bernoulli.

$Y_1, Y_2, \dots, Y_n \sim B(p)$ $\quad p \leftrightarrow \theta$

the pmf of $B(p)$:

$$p(y) = \begin{cases} p & \text{if } y=1 \\ 1-p & \text{if } y=0 \end{cases} \quad \checkmark \checkmark$$

$$p(y) = p^y (1-p)^{1-y} \qquad y = 0, 1$$

$$L(p; y_1, \dots, y_n) = p^{y_1}(1-p)^{1-y_1} \cdot p^{y_2} \cdot (1-p)^{1-y_2} \cdot \dots \cdot p^{y_n}(1-p)^{1-y_n}$$

$$= \prod_{i=1}^n p^{y_i} \cdot \prod_{i=1}^n (1-p)^{(1-y_i)}$$

$$= p^{\sum y_i} (1-p)^{n - \sum y_i} \xrightarrow[p]{} \max$$

For computational reasons, take the $\ln(\cdot)$ (which is increasing) and get the log-likelihood, i.e.,

$$\ell(p; y_1, \dots, y_n) = \sum_{i=1}^{n} y_i \cdot \ln(p) + \left(n - \sum_{i=1}^{n} y_i\right) \cdot \ln(1-p) \xrightarrow[p]{} \max$$

Next, we differentiate with respect to p.

$$\ell'(p; y_1, \dots, y_n) = \sum_{i=1}^{n} y_i \cdot \frac{1}{p} + \left(n - \sum_{i=1}^{n} y_i\right) \cdot \frac{1}{1-p} \cdot (-1)$$

We equate the derivative to zero, and solve for p.

$$\left(\sum y_i\right) \cdot \frac{1}{p} - \left(n - \sum y_i\right) \frac{1}{1-p} = 0 \quad / \cdot p(1-p)$$

$$(1-p)\left(\sum y_i\right) - \left(n - \sum y_i\right) \cdot p = 0$$

$$\sum y_i - p \sum y_i - n \cdot p + p \cdot \sum y_i = 0$$

$$p = \frac{\sum y_i}{n} = \bar{y}$$

$$\boxed{\hat{p}_{MLE} = \bar{Y}}$$

□

Def'n. If $Y_1, \dots, Y_n$ come from a continuous dist'n D w/ pdf f^θ, then the likelihood f'tion is

$$L(\theta; y_1, \dots, y_n) = f^\theta_{Y_1, \dots, Y_n}(y_1, \dots, y_n)$$
$$= f^\theta_{Y_1}(y_1) \cdot f^\theta_{Y_2}(y_2) \cdot \dots \cdot f^\theta_{Y_n}(y_n)$$

Example. **Exponential.**

$Y_1, \dots, Y_n \sim E(\theta)$

the pdf is $f^\theta(y) = \frac{1}{\theta} e^{-\frac{y}{\theta}}$, $y > 0$

$$\Rightarrow L(\theta; y_1, \dots y_n) = \frac{1}{\theta} e^{-\frac{y_1}{\theta}} \cdot \frac{1}{\theta} e^{-\frac{y_2}{\theta}} \cdot \dots \cdot \frac{1}{\theta} e^{-\frac{y_n}{\theta}}$$
$$= \left(\frac{1}{\theta}\right)^n \cdot e^{-\frac{1}{\theta}\sum y_i} \xrightarrow[\theta]{} \max$$

$$\Rightarrow l(\theta; y_1, \dots, y_n) = -n \ln(\theta) + \left(-\frac{1}{\theta}\sum y_i\right) \xrightarrow[\theta]{} \max$$

$$\Rightarrow l'(\theta; y_1, \dots, y_n) = -n \cdot \frac{1}{\theta} + \left(+(+1) \cdot \frac{1}{\theta^2} \cdot \sum y_i\right) = 0 \quad / \theta^2$$

$$-n \cdot \theta + \sum_{i=1}^n y_i = 0$$

$$\hat{\theta} = \bar{y}$$

$$\boxed{\hat{\theta}_{MLE} = \bar{Y}}$$

□

Def'n. An estimator $\hat{\Theta} = \hat{\Theta}(y_1, \dots, y_n)$ is called the **maximum likelihood estimator (MLE)** if it satisfies that for any $\hat{\Theta}' = \hat{\Theta}'(y_1, \dots, y_n)$ we have

$$L(\hat{\Theta}; y_1, \dots, y_n) \geq L(\hat{\Theta}'; y_1, \dots, y_n)$$

for all $y_1, \dots, y_n$

Example. $Y_1, \dots, Y_n \sim U(0, \Theta)$ w/ $\Theta > 0$ unknown.

The pdf of $U(0, \Theta)$:

$$f^{\Theta}(y) = \frac{1}{\Theta} \mathbb{1}_{[0,\Theta]}(y) = \frac{1}{\Theta} \mathbb{1}_{\{0 \leq y \leq \Theta\}}$$

The likelihood:

$$L(\Theta; y_1, \dots, y_n) = \left(\frac{1}{\Theta} \cdot \mathbb{1}_{\{0 \leq y_1 \leq \Theta\}}\right) \cdots \left(\frac{1}{\Theta} \mathbb{1}_{\{0 \leq y_n \leq \Theta\}}\right)$$

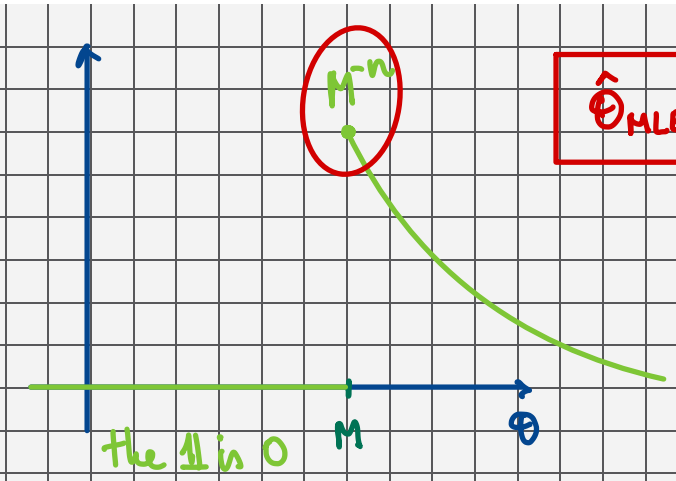
$$= \left(\frac{1}{\Theta}\right)^n \cdot \mathbb{1}_{\{0 \leq y_1, y_2, \dots, y_n \leq \Theta\}}$$

$$= \left(\frac{1}{\Theta}\right)^n \cdot \mathbb{1}_{\{0 \leq \min(y_1, \dots, y_n)\}} \cdot \mathbb{1}_{\{\max(y_1, \dots, y_n) \leq \Theta\}}$$

"assume $y_1, \dots, y_n \geq 0$"

$$= \Theta^{-n} \cdot \mathbb{1}_{\{\max(y_1, \dots, y_n) \leq \Theta\}}$$

$\max(y_1, \dots, y_n) = y_{(n)} = M$



$$\hat{\theta}_{MLE} = \max(Y_1, \ldots, Y_n) = Y_{(n)}$$

Q: Is this the same as moment matching?

→: theoretical mean = sample

$\frac{\theta}{2}$ = $\bar{Y}$

$\Rightarrow$ $\hat{\theta}_{MM} = 2 \cdot \bar{Y}$

□